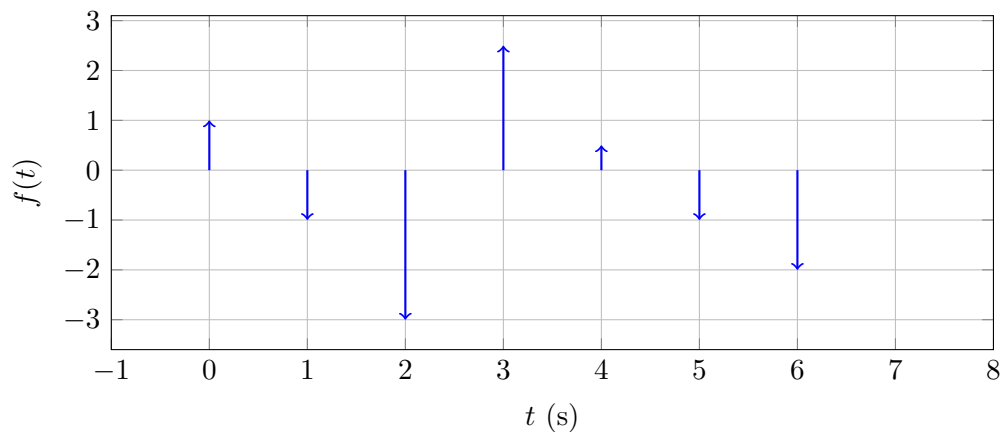


ECE 3210 hw06

1. A zero-order hold reconstructs a signal from the samples

$$\bar{f}(t) = \sum_{n=0}^6 a_n \delta(t - n), \quad (a_0, \dots, a_6) = (1, -1, -3, 2.5, 0.5, -1, -2),$$

using the impulse response $h(t) = \text{rect}(t)$, where $\text{rect}(t) = 1$ for $|t| \leq \frac{1}{2}$ and is zero otherwise. $\bar{f}(t)$ is shown in the figure below with samples taken in 1 second intervals.



- Write the reconstructed signal $f(t)$.
- Determine the frequency response $H(\omega)$ of the hold.
- Find the minimum delay required to make the hold causal.
- Give the delayed reconstructed signal $f_d(t)$.
- Determine the frequency response $H_d(\omega)$ after the delay.

2. Let $f_1(t)$ and $f_2(t)$ be bandlimited signals with bandwidths of 150 Hz and 75 Hz, respectively. Also consider a sinusoid sampled and reconstructed by an ideal low-pass filter, and a first-order-hold reconstruction system.
- (a) Find the Nyquist sampling rate for $f_1(t) - f_2(t)$.
 - (b) Find the Nyquist sampling rate for $f_1^2(t)$.
 - (c) A sinusoid $x(t) = A \sin(2\pi f_0 t)$ with $f_0 = 7700$ Hz is sampled at $f_s = 8000$ Hz and reconstructed with an ideal low-pass filter having cutoff $f_c = f_s/2$. What tone frequency is heard?
 - (d) A first-order hold with sampling interval $T = 1$ s has impulse response $h(t) = \Delta(t/2)$, where $\Delta(u) = \max(1 - 2|u|, 0)$. The samples are $\tilde{f}(t) = \sum_{n=-1}^6 a_n \delta(t - n)$, with $(a_{-1}, \dots, a_6) = (0, 1, 2, 1, -1, -2, -3, 1)$. Write the reconstructed signal $f(t)$ and sketch it along with the sample values.
 - (e) Determine the frequency response $H(\omega)$ of the first-order hold.
 - (f) Find the minimum delay required to make the filter realizable, and give the resulting frequency response $H_d(\omega)$.
 - (g) How does the delay affect the reconstructed signal?

3. Consider the signal

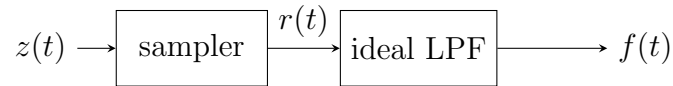
$$f(t) = \frac{1}{\pi} \operatorname{sinc}^2\left(\frac{t}{2}\right).$$

- (a) Sketch its Fourier transform $F(\omega)$.
- (b) Find the signal bandwidth in hertz and the lowest sampling rate that avoids aliasing.
- (c) Suppose the signal is sampled at one-half the Nyquist rate. Find the Fourier transform $\bar{F}(\omega)$ of the sampled signal.
- (d) Find an expression for $\bar{f}(t)$, the sampled signal in the time domain (using delta functions).
- (e) Reconstruct the sampled signal using an ideal low-pass filter with cutoff frequency one-quarter of the Nyquist sampling rate and unity passband gain. Find the resulting signal $f_{\text{recon}}(t)$.

4. Consider the message signal

$$f(t) = \frac{1}{2\pi} \operatorname{sinc}^2\left(\frac{t}{4}\right)$$

which is modulated by a carrier according to $z(t) = f(t) \cos(2t)$. The receiver samples $z(t)$ and then applies an ideal low-pass filter to recover $f(t)$.



- (a) Determine the Fourier transform $F(\omega)$.
- (b) Determine the Fourier transform $Z(\omega)$ of the modulated signal.
- (c) For a sampling rate $f_s = 1/\pi$ Hz, find the sampling period T_s , sampling angular frequency ω_s , and the spectrum $R(\omega)$ of the sampled signal.
- (d) What is the largest sampling period T that still permits recovery of $f(t)$ using this receiver?

5. Consider the bandlimited signal

$$f(t) = \frac{1}{\pi} \operatorname{sinc}^2\left(\frac{t}{2}\right)$$

- (a) Determine the Fourier transform $F(\omega)$.
- (b) Find the bandwidth in hertz and the Nyquist sampling rate.
- (c) The signal is sampled at three-quarters of the Nyquist sampling rate. Plot the Fourier transform of the sampled signal $\bar{F}(\omega)$.
- (d) Instead, modulate the original signal by $\cos(0.5t)$, so that $f_m(t) = f(t) \cos(0.5t)$. Determine $F_m(\omega)$.

6. Consider the complex-valued signal

$$f(t) = \frac{1}{2} \operatorname{sinc}^2\left(\frac{\pi t}{2}\right) e^{j\pi t}$$

- (a) Derive the Fourier transform $F(\omega)$.
- (b) Using the highest frequency represented in the signal as its bandwidth, find the bandwidth B in hertz.
- (c) Find the longest sampling period $T_{\max}$ that avoids aliasing and the corresponding sampling rate f_s .
- (d) Sketch the spectrum $\bar{F}(\omega)$ of the sampled signal and state whether the original signal can be recovered from these samples.

Answer Key

- 1(a) $f(t) = \sum_{n=0}^6 a_n \text{rect}(t - n)$, with the listed a_n .
- 1(b) $H(\omega) = \text{sinc}(\omega/2)$.
- 1(c) $T_d = \frac{1}{2}$ s.
- 1(d) $f_d(t) = f(t - \frac{1}{2})$.
- 1(e) $H_d(\omega) = e^{-j\omega/2} \text{sinc}(\omega/2)$.
- 2(a) $f_{\text{Nyq}} = 300$ Hz.
- 2(b) $f_{\text{Nyq}} = 600$ Hz.
- 2(c) 300 Hz.
- 2(d) $f(t) = \sum_{n=-1}^6 a_n \Delta((t - n)/2)$.
- 2(e) $H(\omega) = \text{sinc}^2(\omega/2)$, where $\text{sinc}(x) = \sin(x)/x$.
- 2(f) $T_d = 1$ s and $H_d(\omega) = e^{-j\omega} \text{sinc}^2(\omega/2)$.
- 2(g) $f_d(t) = f(t - 1)$.
- 3(a) $F(\omega) = 2 \max(1 - |\omega|, 0)$.
- 3(b) $B = \frac{1}{2\pi}$ Hz and $f_{\text{Nyq}} = \frac{1}{\pi}$ Hz.
- 3(c) $\bar{F}(\omega) = 1/\pi$.
- 3(d) $\bar{f}(t) = \frac{1}{\pi} \delta(t)$.
- 3(e) $f_{\text{recon}}(t) = \frac{1}{2\pi^2} \text{sinc}(t/2)$.
- 4(a) $F(\omega) = 2\Delta(\omega)$.
- 4(b) $Z(\omega) = \Delta(\omega - 2) + \Delta(\omega + 2)$.
- 4(c) $T_s = \pi$, $\omega_s = 2$, and $R(\omega) = \frac{1}{\pi} \sum_{k \in \mathbb{Z}} Z(\omega - 2k)$.
- 4(d) $T_{\text{max}} = 2\pi$ s.
- 5(a) $F(\omega) = 2\Delta(\frac{\omega}{2})$.
- 5(b) $B = \frac{1}{2\pi}$ Hz and $f_{\text{Nyq}} = \frac{1}{\pi}$ Hz.
- 5(c) $\bar{F}(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} F(\omega - k\omega_s) = \frac{1.5}{2\pi} \sum_{k=-\infty}^{\infty} F(\omega - k(1.5))$.
- 5(d) $F_m(\omega) = \frac{1}{2} [F(\omega - 0.5) + F(\omega + 0.5)]$.
- 6(a) $F(\omega) = \Delta(\frac{\omega - \pi}{2\pi})$.
- 6(b) $B = 1$ Hz.
- 6(c) $T_{\text{max}} = 1$ s and $f_s = 1$ Hz.
- 6(d) $\bar{F}(\omega) = \sum_{k \in \mathbb{Z}} F(\omega - 2\pi k)$. Yes.